

Bishal Marasini

Aerospace Engineer

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Research Interests

Focused on aircraft design, aircraft performance, and flight dynamics, with an emphasis on aerodynamic analysis and performance optimization. Research also explores CFD, aircraft manufacturing, UAV autonomy, control systems, and structural analysis to improve the efficiency, stability, and overall performance of aerospace vehicles.

Education

Bachelor's Degree in Aerospace Engineering

2022–2026

- *Pulchowk Campus, Institute of Engineering, Tribhuvan University, Nepal*
- Percentage: 77.42%

Relevant Coursework: *Flight Dynamics, Control Systems, UAV, Aircraft Preliminary Design, Aerodynamics, Fluid Dynamics, CFD, FEM, Strength of Materials, Propulsion, Aircraft Structures, CAD & Manufacturing*

Research and Professional Experience

Research Intern, FOSSEE, IIT Bombay

Ongoing

- **Topic:** *CFD Simulation of Supersonic Intake of a Scramjet Engine*
- Developing the geometry and computational domain and generating and refining the mesh for a supersonic scramjet intake.
- Setting up compressible-flow simulations in OpenFOAM, including solver selection, to investigate flow behaviour inside the intake.

Trainee Performance Engineer, Operations Department, Yeti Airlines

Sept 2025–Mar 2026

- Performed aircraft performance calculations for takeoff, landing, and climb under varying weight and environmental conditions.
- Supported flight planning, fuel estimation, payload optimization, and operational performance analysis using aircraft flight manuals and operational constraints.
- Assisted with aircraft weight and balance calculations, including load and trim sheets and assessment of last-minute operational changes.

Publications

B. Karki, **B. Marasini**, P. Baniya, and S. Baskota, “Crop Health Monitoring Through Aerial Mapping Using Quadcopter Based Imaging,” **Thesis**. Available: TU Repository

Academic Projects

Crop Health Monitoring Through Aerial Mapping Using Quadcopter Based Imaging [THESIS] 2025-2026

- Developed a quadcopter based aerial imaging and geospatial mapping workflow for crop health monitoring of affected regions.
- Applied computer vision and Random Forest machine learning for crop-health classification and georeferenced mapping.
- Developed a computational workflow integrating UAV imagery, machine learning, and geospatial analysis.
- **Tools:** *Pix4D, Python, Drone Deploy, QGIS*

Design and Analysis of Sky-Yak: 19-Seat STOL Aircraft [Poster Presentation]

Aug 2025

- Carried out sizing, aerodynamic analysis, performance evaluation, and stability and flight-dynamics analysis for a 19-seat STOL aircraft.

- Built and evaluated aircraft models in XFLR5, Plane Maker, X-Plane, and MATLAB.
- Conducted simulator-based flight tests to assess V-n characteristics, CG travel, stall behavior, take-off/landing performance, and handling qualities.
- **Tools:** *XFLR5, Plane Maker, X-Plane, MATLAB*

Design and Analysis of Hawk-Eye: Wildlife Surveillance UAV **Mar 2025**

- Performed UAV sizing, performance, stability, and dynamic analysis for a wildlife-surveillance mission.
- Evaluated performance and stability characteristics through simulator-based flight tests.
- **Tools:** *XFLR5, Plane Maker, X-Plane, MATLAB*

Design and Fabrication of Small-Scale Wooden Aircraft Model **Feb 2025**

- Designed an aircraft model in CATIA considering aerodynamic and structural requirements.
- Performed aerodynamic analysis in XFLR5 to evaluate lift, drag, and stability characteristics.
- Fabricated a wooden prototype based on the CAD model.
- **Tools:** *CATIA, XFLR5*

Study of Aerodynamic Forces on a NACA 0018 Foil Using DAQ System in a Low Subsonic Wind Tunnel **Mar 2024**

- Measured lift on a NACA 0018 airfoil in an open-loop subsonic wind tunnel using load cells for lift and drag with a DAQ system, to study the variation of aerodynamic coefficients with angle of attack and flow velocity.
- Computed lift coefficient versus angle of attack and examined the linear region and stall onset, which was associated with flow separation.
- Assessed measurement and calibration errors, including a non-zero lift coefficient at zero angle of attack.
- **Tools:** *Wind tunnel, load cells, DAQ system*

Research Projects

CFD Simulation of Supersonic Intake of a Scramjet Engine **Sept 2026 -Present**

- Simulated Mach 5 flow through a 2D multi-block ramp-cowl intake in OpenFOAM, using rhoCentralFoam with the $k-\omega$ SST turbulence model.
- Generated structured multi-block meshes, controlling non-orthogonality and skewness, and ran cases in parallel.
- Working toward a comparison of compressible-flow solvers for shock-dominated supersonic intake flow.
- **Tools:** *OpenFOAM, ParaView, Python*

Airfoil Optimization Using Genetic Algorithm, SIC Hackathon **May 2025**

- Optimized airfoil performance using a genetic algorithm coupled with XFOIL.
- Combined aerodynamic analysis, programming, and optimization techniques.
- **Tools:** *Python, XFOIL*

Autonomous Delivery Using Quadcopter **Dec 2024**

- Integrated Pixhawk 4 and PX4 for autonomous waypoint navigation and mission execution.
- Developed a servo-actuated payload release mechanism triggered through QGroundControl mission commands.
- Conducted autonomous flight tests and analyzed flight logs, actuator response, trajectories, and velocity profiles.
- **Tools:** *PX4, QGroundControl*

Courses and Training

ATR-72 Aircraft Performance Weight and Balance Training, <i>Yeti Airlines Ltd.</i>	Oct 2025
Training on Dangerous Goods Regulations, <i>Yeti Airlines Ltd.</i>	Oct 2025
Training on Human Factor in Aviation, <i>Yeti Airlines Ltd.</i>	Oct 2025
Training on Aviation Security and Safety Management System (SMS), <i>Yeti Airlines Ltd.</i>	Oct 2025
Coding and Programming, <i>Samsung Innovation Campus</i>	May 2025

Professional and Technical Activities

Member, Flying Club, Pulchowk Campus **2024-Present**

- Contributed to UAV fabrication, system integration, and flight testing.
- Collaborated with aerospace engineering teams on hands-on UAV design and development projects.
- Conducted autonomous flight operations and evaluated UAV flight performance.

Event Coordinator, National Mechanical and Aerospace Exhibition, Pulchowk Campus **Jan 2025**

- Coordinated an engineering-student event (RC car race) and technical workshops.
- Organized technical competitions and the MechTRIX exhibition.

Volunteer, National Mechanical and Aerospace Exhibition, Pulchowk Campus **Feb 2023**

- Volunteered at events including Robo-soccer and Go-kart race.

Key Competencies

Programming: Python, MATLAB, C

CFD: OpenFOAM (rhoCentralFoam, simpleFoam, snappyHexMesh), ANSYS Fluent, ParaView

UAV & Autonomy: PX4, Pixhawk, QGroundControl

CAD & Design: CATIA, XFLR5, X-Plane, Plane Maker

Optimization & ML: Genetic Algorithms, Random Forest, Computer Vision

Geospatial & Data: QGIS, Pix4D, DroneDeploy

Other: Microsoft Excel, Microsoft Office, LaTeX

Organizational Affiliation

Nepal Engineering Council (NEC) **Oct 2026**

- General Registered Engineer - Aerospace Engineering

References

Kamal Darlami, Assistant Professor, Former Deputy Head of Department

Department of Mechanical and Aerospace Engineering, Pulchowk Campus, IOE, Tribhuvan University

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Currently pursuing PhD at the University of Southern Queensland

Biman Rimal, Assistant Professor, Deputy Head of Department

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